ELI — Diagrams

ELI v2.5.16

October 2026

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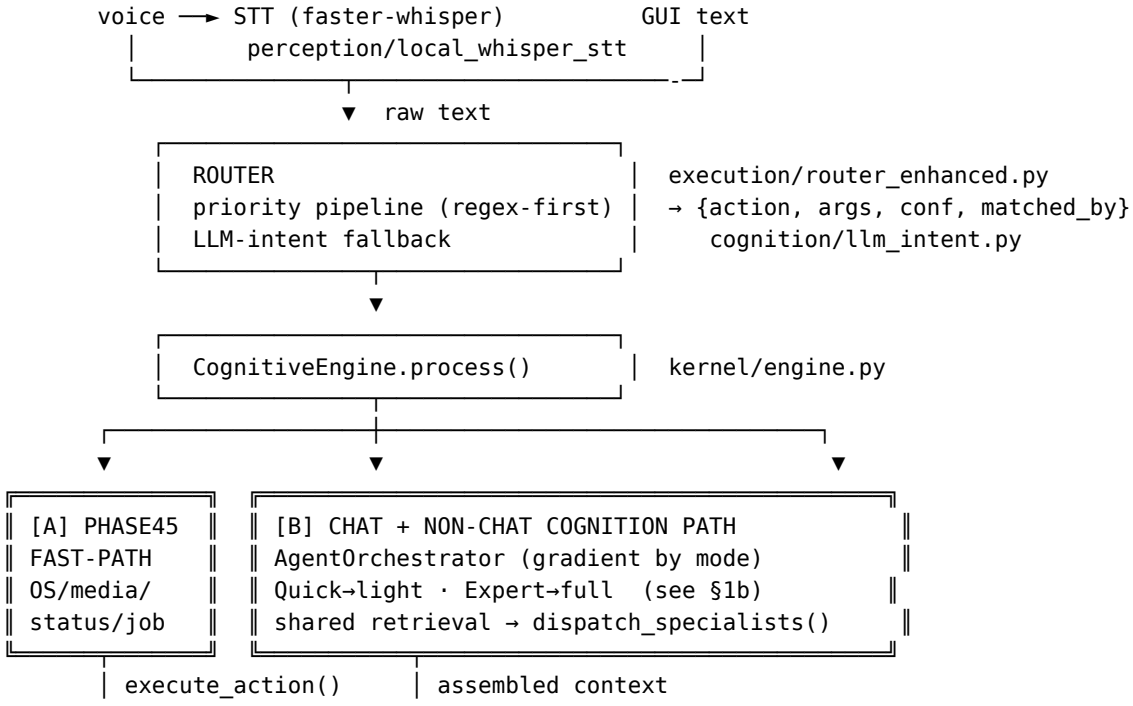
Blueprint — ELI ASCII Diagrams

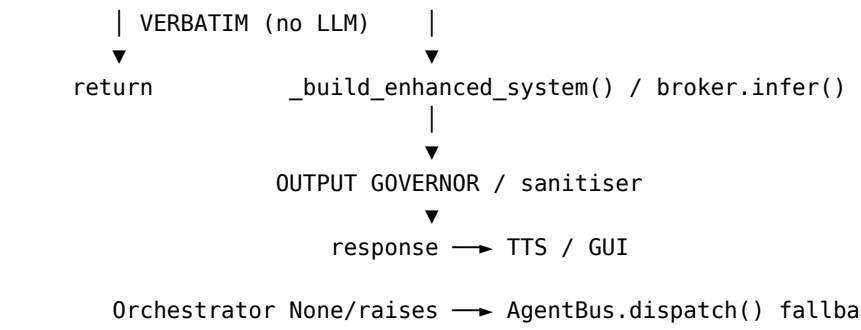
> **Updated for v2.4.84.** CHAT uses a gradient orchestrator for all modes; retrieval is unified in `eli/memory/retrieval.py`; Stage 12 = learning/state commit.

Visual companion to `architecture.md`. Three views: the full request pipeline, the memory subsystem, and the gating stack. All grounded in the real modules (file paths are clickable).

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1. Full pipeline (request lifecycle)



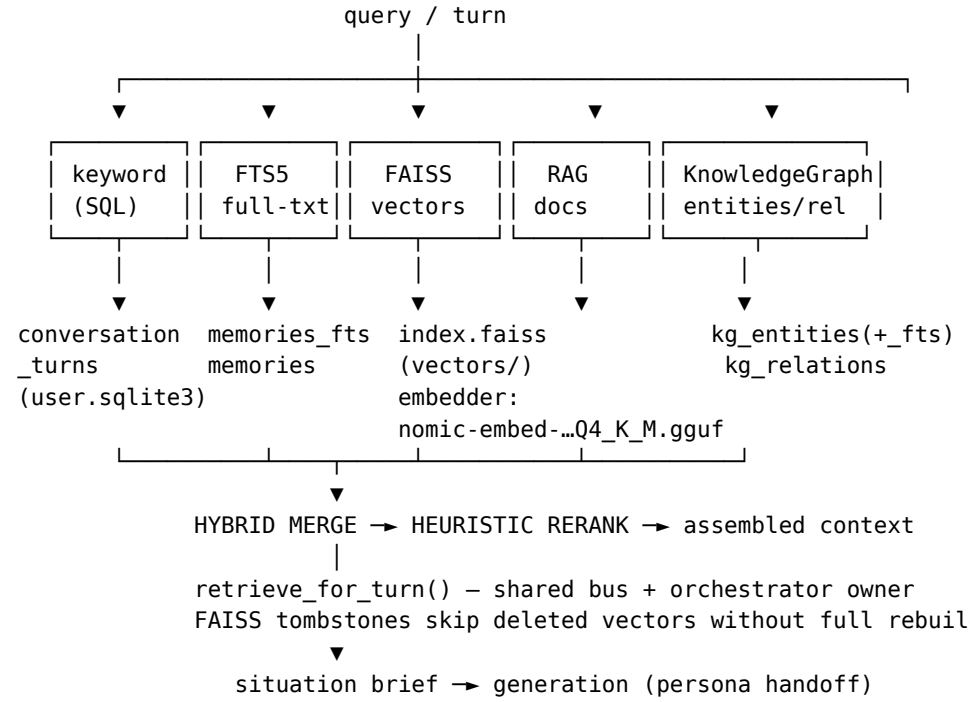


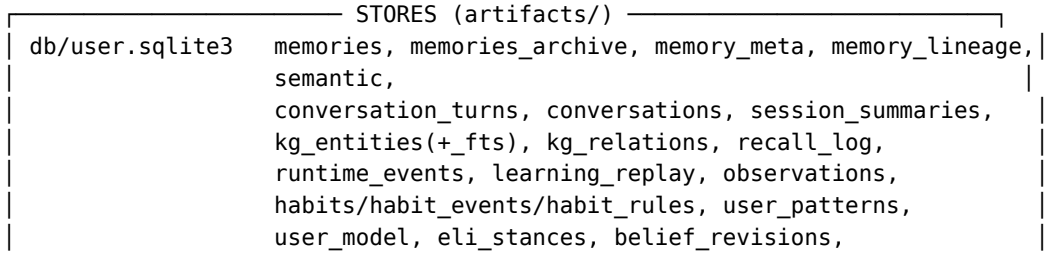
1b. Orchestrator — canonical S01–S12 (`pipeline_trace.py`)

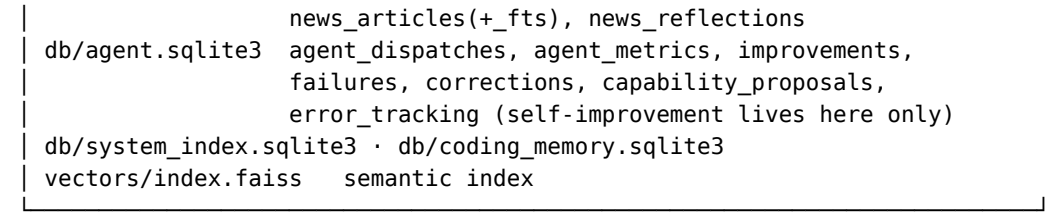
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S01 PERCEIVE → S02 INPUT_GUARDS → S03 ROUTER → S04 GROUNDING_GATE
       → S05 PLANNER (mode-aware budgets)
       → S06 AGENT_BUS: retrieve_for_turn() + dispatch_specialists()
       → S07 CONTEXT_ASSEMBLY (merge + heuristic rerank)
       → S08 INFERENCE_BROKER → S09 REASONING_SYNTHESIS (mode algo)
       → S10 OUTPUT_GOVERNOR → S11 RESPONSE_DELIVERY
       → S12 LEARNING_STATE_UPDATE (learning_coordinator.finalize_turn)
```

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2. Memory subsystem `eli/memory` + `cognition/working_memory.py`





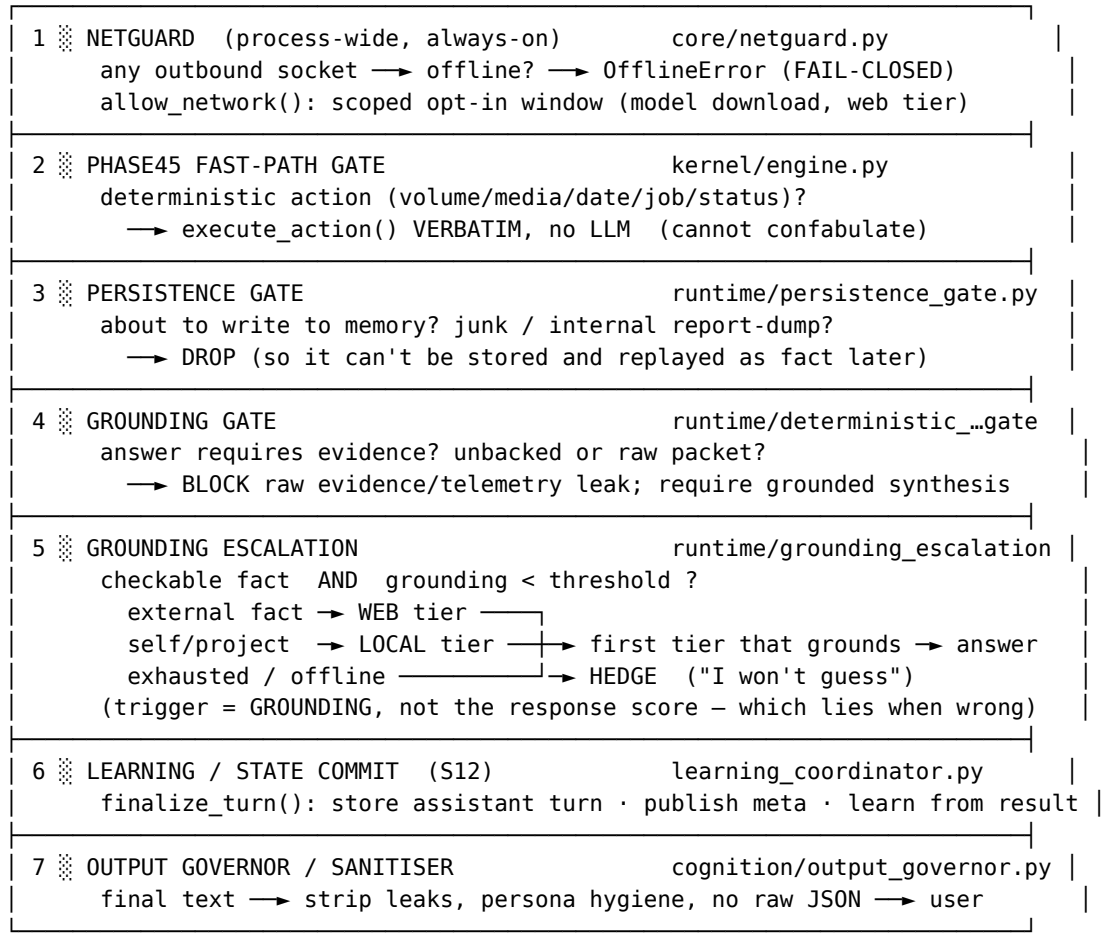


A question that names a period ("last week") is filtered by date FIRST (Memory.memories_between + dated conversation turns), then ranked by topic.

WRITE PATH: turn → PERSISTENCE GATE (drop junk/report-dumps) → STORAGE POLICY (origin · dedupe · importance) → store; daily upkeep decays and archives

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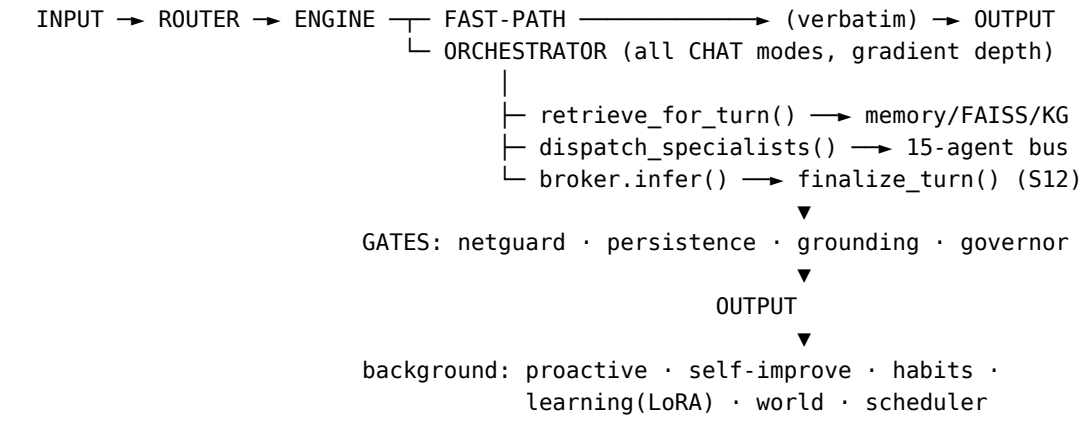
3. Gating stack (in path order — every guard the request passes)



Legend: FAIL-CLOSED = denies by default; VERBATIM = returned without an LLM pass; HEDGE = honest "can't verify" instead of a guess.

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4. One-screen overview (how it all connects)



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